Household Behavior over the Life Cycle

Assignment 1: Labor Supply and Children

lhr250

1. *What value of β_1 produces an event-study graph of working hours with something close to a drop in hours of 10pct around the time of birth as in Kleven, Landais and Søgaard (2019, Fig. 1B)? Be explicit about how you found the value.*

   In the model the disutility of working is scaled by the child-penalty function $\beta(n_t) = \beta_0 + \beta_1 n_t$, so that the presence of a child ($n_t = 1$) raises the marginal disutility of working. β_1 is therefore the "child penalty" parameter: A higher β_1 makes working more costly once a child arrives and the individual optimally responds by reducing working hours. This is the mechanism that generates the drop in labor supply around the time of birth.

   To find the correct β_1 value I use the structural estimation logic from lecture 2. I define the moment $m(\beta_1)$ as the percentage change in average simulated hours at the time of birth (event time 0) relative to the period before birth (event time - 1), and solve for the β_1 that makes the model-implied drop equal to the -10% target: $m(\beta_1) = -0.10$.

   With only one target ($m(\beta_1) = -0.10$) and one parameter (β_1), I can solve for β_1 directly instead of minimizing a distance. I use a root-finder (scipy.optimize.brentq) on $\beta_1 \in [0, 0.5]$, which solves and simulates the model for different values of β_1 until the drop in hours is -10%. A larger β_1 always results in a larger drop in working hours (and no drop at $\beta_1 = 0$), so exactly one value fits. Since the random draws for child arrival are fixed, I will get the same answer every time I run the code.

   This results in a $\hat{\beta}_1 = 0.0558$. Figure 1 shows the event-study graph of hours worked relative to the period before birth. This illustrates the drop in hours of 10 pct immediately after the child is born, and afterwards increasing again as the child grows older.

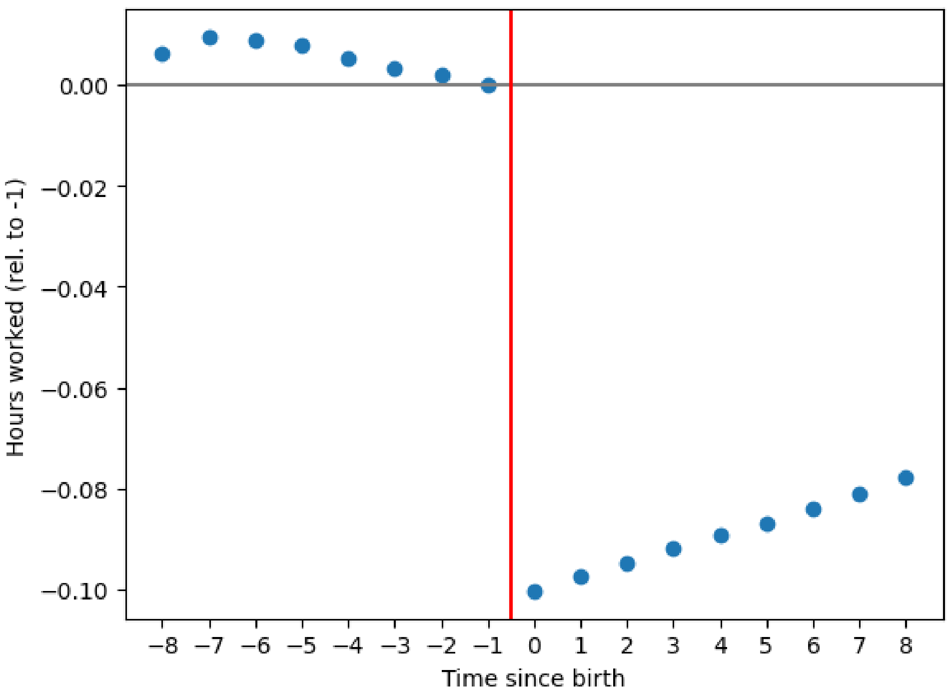


Figure 1: Hours worked relative to the period before birth

2. *Introduce in the model a spouse who contributes with income y_t every period. Let the income depend on age/period, such that $y_t = 0.1 + 0.01 \cdot t$. Write out the affected equations and implement it in Python. Compare and comment on the simulated behavior from this alternative model in relation to that of the baseline model above. Concretely plot age profiles of e.g. labor supply, consumption, fertility and savings.*
   [$y_t = 0\ \forall t$ *gives the baseline model.*]

   The only equation affected by adding a spouse to the model is the budget constraint. The spouse contributes with an exogenous income y_t which is added to the household's resources every period independently of own labor supply. The income y_t is added to the budget constraint a_{t+1} as shown below:

$$a_{t+1} = (1 + r)(a_t + (1 - \tau)w_t h_t + y_t - c_t)$$

   Where $y_t = 0.1 + 0.01 \cdot t$.

   We notice that the spouse earns a bit more each year (from 0.1 when $t = 0$ to 0.19 in the last period $t = 9$), so the household gets gradually richer with age.

   Figure 2 compares the two models and how consumption (c), savings (a), labor supply (h) and fertility (n) is affected when a spouse is included in the model. The intuition for each profile is:

   - With a spouse present consumption is higher in all periods, which makes sense since the household has more money to spend. The increase is fairly constant across the life cycle, which can reflect consumption smoothing.
   - The savings are almost identical in both models. Both borrow when young (wages are low early, so they borrow to smooth consumption and repay later) and the spouse's income barely shifts the pattern as the extra income is mostly consumed rather than saved. The small difference reflects the fact that early and mid-life the spouse's income is still low, while consumption is already raised (consumption smoothing), so the household borrows marginally more. Later in life the spouse's income increases which results in the household having marginally more wealth.
   - The labor supply falls as the household is richer with a spouse present. Reflects an income effect as the extra income from the spouse means that you don't have to work as much yourself. There is no substitution effect as the wage remains unaffected.
   - Fertility is exactly the same with and without a spouse. Children arrive randomly with a fixed probability that has nothing to do with income or whether a spouse is around. In reality this is a limitation of the model, as having a partner and more income is likely to affect the decision of having children.

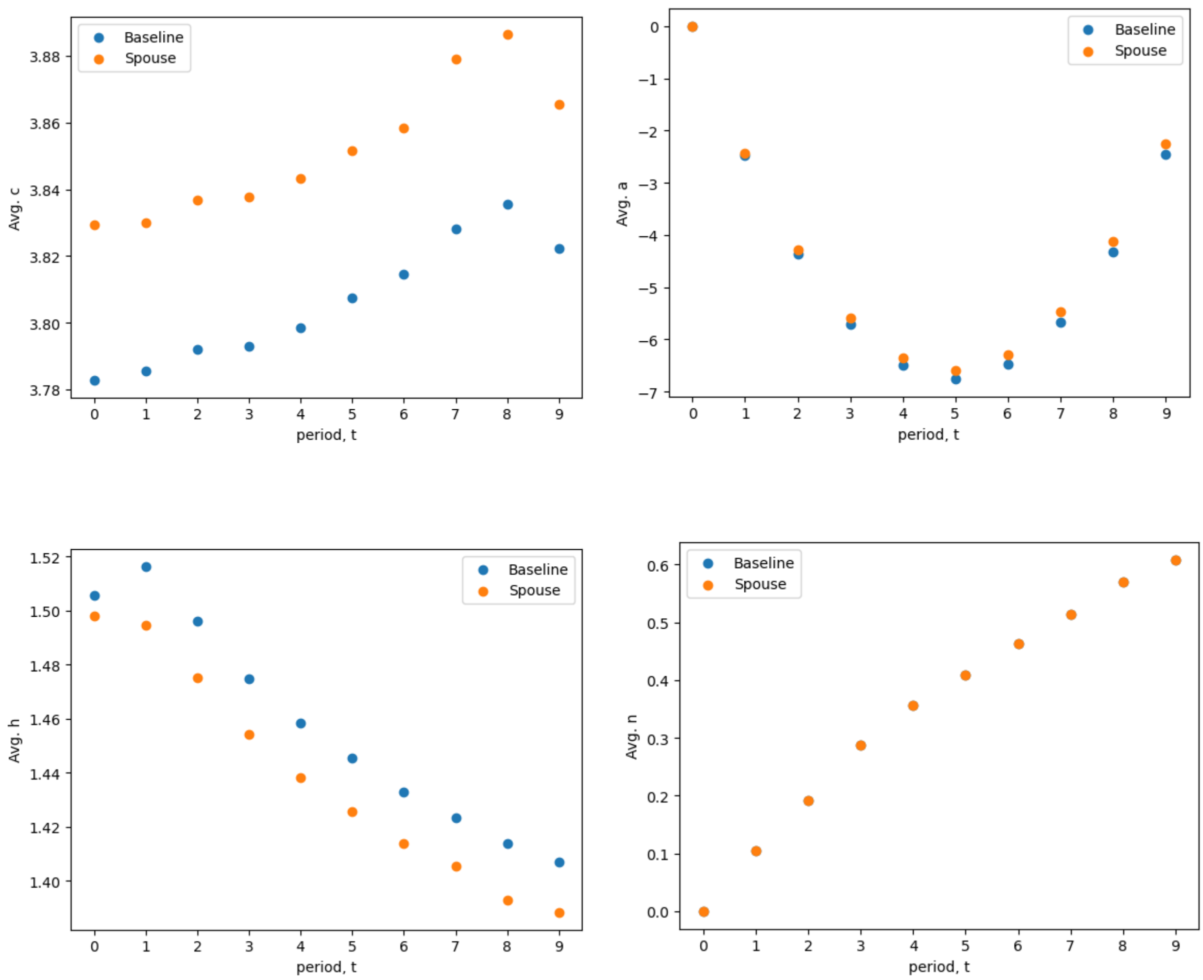


Figure 2: Baseline and spouse model

3. *Introduce in the model a childcare cost of $\theta = 0.05$ in the presence of a child. You can think of this as a policy reform reducing the child care subsidy. Write out the affected equations and implement it in Python. Compare the simulated behavior from this alternative model in relation to that of the baseline model above. Compare also the Marshall elasticity in both models and comment on how the income and substitution effects differs (you do not need to calculate them).*
   [$y_t = 0\ \forall t$ *and* $\theta = 0$ *gives the baseline model.*]

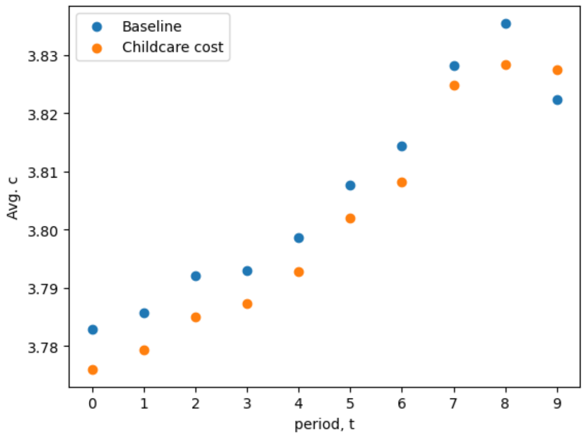
   Only the budget constraint is affected. The childcare cost is subtracted from the household's resources whenever a child is present:

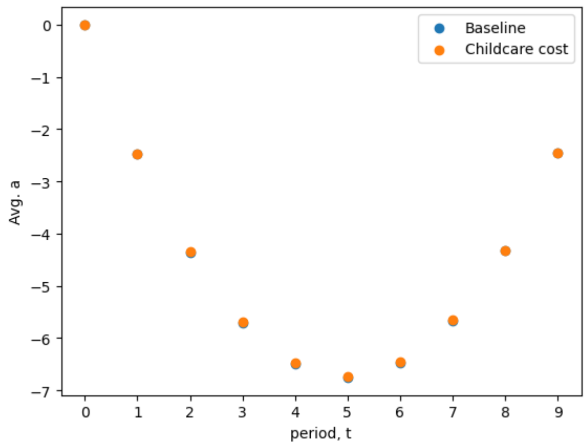
$$a_{t+1} = (1 + r)(a_t + (1 - \tau)w_t h_t + y_t - c_t - \theta \cdot \mathbf{1}\{n_t > 0\})$$

   The spouse income y_t is retained in the budget constraint, but for this experiment it is switched off ($y_t = 0$) so that the comparison isolates the effect of the childcare cost. This matches the assignment's baseline, which sets $y_t = 0$ and $\theta = 0$.

Figure 3 illustrates how consumption (c), savings (a), labor supply (h) and fertility (n) is affected when a childcare cost is present.

- Consumption is lower with the childcare cost present in every period (except from the last at t=9). The extra cost of taking care of the child θ reduces the household's disposable income, so they can afford less.
- Savings are essentially identical in the two models. The childcare cost is small ($\theta = 0.05$) and is only paid once a child is present, so it is a small reduction in the household's resources. It is absorbed mainly through slightly lower consumption (and slightly more work), which leaves the borrowing/saving path almost unchanged.
- The labor supply is slightly higher with the childcare cost and child present. This is an income effect as the cost lowers the household's resources without changing the wage/reward for an hour of work, so there is no substitution effect. The household is simply poorer and works a little more to make up for the lost resources.
- Fertility is identical in the two models. Children arrive randomly with a fixed probability and the process uses the same random draws, so the childcare cost cannot change the fertility by construction. In reality a higher cost of children would likely reduce fertility, which reflects a limitation of the model.





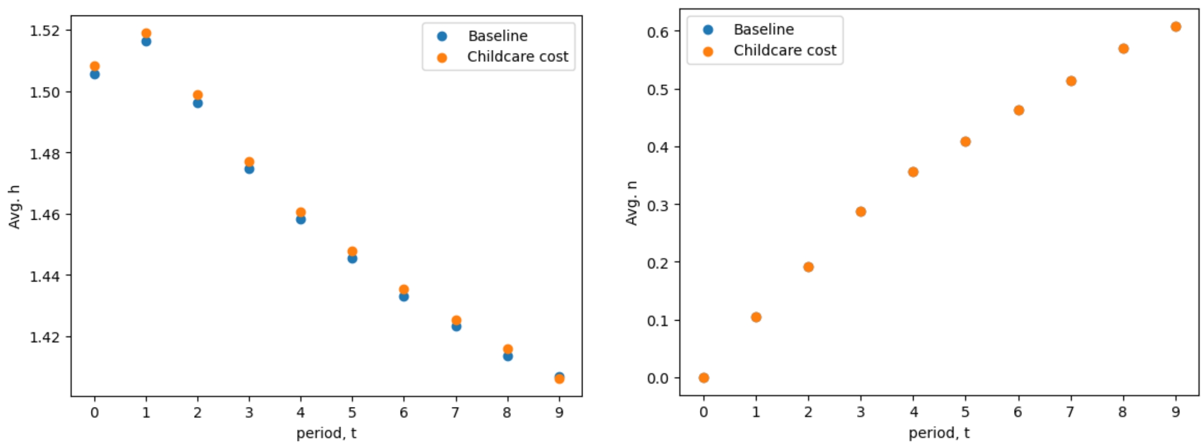


Figure 3: Baseline and childcare cost model

The Marshall elasticity measures how average hours respond to a 1% rise in wages, combining the substitution effect (a higher wage makes each hour of work more rewarding, so people work more) and the income effect (a higher wage makes the household richer, so it wants more leisure and works less). We find an elasticity of -0.197 in the baseline and -0.196 with the childcare cost. Both are negative, meaning the income effect slightly outweighs the substitution effect: as the wage increase is permanent, the household becomes wealthier so it prefers to have more leisure and thus work less. The two values are almost identical because the childcare cost is small and only paid when a child is present, so it does not affect the Marshall elasticity very much.

4. *Discuss how endogenous fertility adjustments could affect the impact of a child subsidy reform as the one above. Specifically, think about substitution and income effects related to fertility.*

   In our model fertility is exogenous: children arrive with a fixed probability that does not depend on θ, so the model captures only the direct channel. The reform above reduces the childcare subsidy by introducing a childcare cost, which makes having a child more expensive. If fertility was endogenous, the household would choose how many children to have, and the reform changes that decision through two effects. The substitution effect states that children are now more expensive relative to consumption, so the household shifts away from children which results in a decline in fertility. According to the income effect, the reform results in the household becoming poorer, and if a child is a normal good, then a lower income reduces the demand for children resulting in fertility declining. Both effects push in the same direction resulting in a decline in fertility.

   There are two effects on labor supply. The direct effect states that introducing a childcare cost makes children more expensive and leaves the household poorer. This results in the household working more to make up for the lost resources (income effect from question 3). As we found above, introducing the reform will result in the fertility declining. With fewer children the child penalty is lower: $\beta(n_t) = \beta_0 + \beta_1 n_t$, so working is less costly and the household works more. Both of these effects push labor supply up.

As the model holds fertility fixed, it only captures the direct effect and ignores the additional rise in labor supply resulting from the households choosing to have fewer children. Since the two channels work in the same direction, the model underestimates how much the reform raises labor supply.

5. *Imagine that whether an income-generating spouse is present or not is random and* $p_s = 0.8$ *is the likelihood that a spouse is present in a given period. Furthermore, imagine that a child can only arrive next period if a spouse is currently present. Formulate mathematically the recursive formulation of this model and implement it in Python. Compare the simulated behavior from this alternative model in relation to that of the baseline model above.*
[$y_t = 0\ \forall t,\ \theta = 0$ *and* $p_s = 1$ *gives the baseline model.*]

We add one state variable, which is whether an income generating spouse is present or not $s_t \in 0, 1$ with probability $p_s = 0.8$. As s_t is a state, it enters the value function on both sides of the Bellman equation is:

$$V_t(s_t, n_t, a_t, k_t) = \max_{c_t, h_t} \frac{c_t^{1+\eta}}{1+\eta} - \beta(n_t)\frac{h_t^{1+\gamma}}{1+\gamma} + \rho\mathbb{E}_t[V_{t+1}(s_{t+1}, n_{t+1}, a_{t+1}, k_{t+1})]$$

We now add the spouse income to the budget constraint. The spouse income y_t only enters when a spouse is present, $s_t = 1$:

$$a_{t+1} = (1+r)(a_t + (1-\tau)w_t h_t + s_t y_t - c_t - \theta \cdot \mathbf{1}\{n_t > 0\})$$

For this experiment $\theta = 0$, the baseline additionally sets $y_t = 0\ \forall t$ and $p_s = 1$. The childcare cost is therefore not removed from the model, but is simply "switched off" in this experiment, exactly as in question 3.

A child can only arrive next period if a spouse is present this period, so the birth probability is conditioned on the current spouse state. The probability of a child arriving now looks like:

$$p(n_t, s_t) = \begin{cases} p_n & \text{if } n_t = 0 \text{ and } s_t = 1 \\ 0 & \text{otherwise} \end{cases}$$

The spouse state is drawn independently each period:

$$s_{t+1} = \begin{cases} 1 & \text{with probability } p_s \\ 0 & \text{with probability } 1 - p_s \end{cases}$$

Calculating the expected future value we need to take into account whether a spouse is present (s_{t+1}) and if a child is arriving (n_{t+1}). We have the following possible outcome combinations:
If a spouse is present next period (with probability p_s):

- Child arriving with probability $p(n_t, s_t) \rightarrow V_{t+1}(s' = 1, n_t + 1, a_{t+1}, k_{t+1})$

- No child arriving with probability $1 - p(n_t, s_t) \to V_{t+1}(s' = 1, n_t, a_{t+1}, k_{t+1})$

If a spouse is not present next period (probability $1 - p_s$):

- Child arriving with probability $p(n_t, s_t) \to V_{t+1}(s' = 0, n_t + 1, a_{t+1}, k_{t+1})$
- No child arriving with probability $1 - p(n_t, s_t) \to V_{t+1}(s' = 0, n_t, a_{t+1}, k_{t+1})$

The above combinations is then weighted with each of their probabilities which can be written as.

$$\mathbb{E}_t[V_{t+1}] = p_s\, p(n_t, s_t)\, V_{t+1}(1, n_t + 1, a_{t+1}, k_{t+1}) + p_s\, (1 - p(n_t, s_t))\, V_{t+1}(1, n_t, a_{t+1}, k_{t+1})$$

$$+ (1 - p_s)\, p(n_t, s_t)\, V_{t+1}(0, n_t + 1, a_{t+1}, k_{t+1}) + (1 - p_s)\, (1 - p(n_t, s_t))\, V_{t+1}(0, n_t, a_{t+1}, k_{t+1})$$

The model is implemented in the same DynLaborFertModel.py as question 1-4 by adding a spouse dimension to the solution arrays and a loop over $s_t \in \{0, 1\}$ in solve, with $p_s = 1$ as the default. With $p_s = 1$ the model collapses to the baseline, so question 1-4 are numerically unchanged (the event-study estimate is still $\beta_1 = 0.0558$). Only question 5 sets $p_s = 0.8$ and $y_t > 0$. Spouse presence and child arrival are drawn once from fixed and seeded arrays so the simulated path is identical on every run.

Figure 4 compares the two models and how consumption (c), savings (a), labor supply (h) and fertility (n) is affected when a random, income-generating spouse is included in the model.

- Consumption is higher than the baseline in every period. The spouse brings in extra income when present. As there is a smaller probability that a spouse is present, children are less likely, so the household expects a smaller "child penalty" on utility. With the budget relaxed and the expected cost of children reduced, the household optimally consumes more at every age.
- Savings are almost the same as the baseline. The extra spouse income is mostly spent rather than saved, and the small change in fertility barely affects how much the household sets aside. The result is roughly the same path, borrowing early and slightly higher assets late.
- Labor supply is a bit lower than the baseline in every period. Two forces pull in opposite directions: the spouse's income makes the household richer, so it wants to work less, while having fewer children lowers the disutility of working, which would make it work more. The income effect is stronger, so the working hours end up slightly lower. The gap is largest early and mid-life, when the spouse's income matters most and gets smaller near the end (except for the last period).
- Fertility is lower than the baseline because births are dependent on a spouse being present, which only happens with a probability of $p_s = 0.8$. So the chance of a birth each period is smaller and fewer children are born over the life cycle.

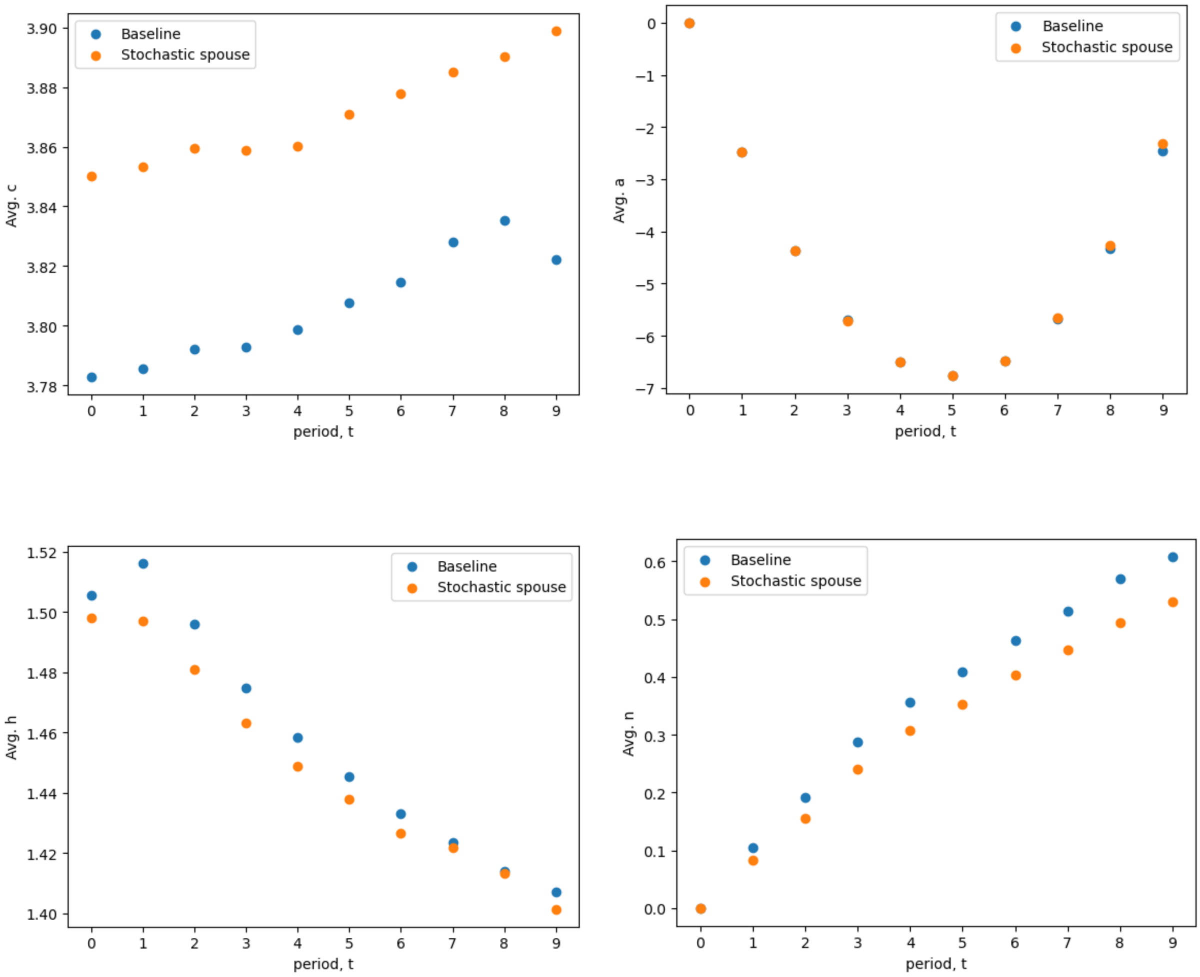


Figure 4: Baseline and stochastic spouse model